



11605 - Calibrating 3.3 micron PAH feature as a star-formation indicator at cosmic noon

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Par028	MIRI Wide Field Slitless Spectroscopy	(1) Par028
	2	Par052	MIRI Wide Field Slitless Spectroscopy	(2) Par052

ABSTRACT

Dust in galaxies is intrinsically tied to star-formation as it reprocesses the light from stars into IR emission. Polycyclic aromatic hydrocarbons (PAHs) are dust grains found ubiquitously in galaxies and have been widely used as tracers of star-formation. However, it is unclear if empirical calibrations from the local universe will continue to hold at earlier epochs, where galaxy properties differ significantly.

We propose a comprehensive, cross-wavelength survey of PAH-3.3 micron emission, the ideal tracer for dust-obscured star-formation at $z > 0.5$, using the newly available observing mode MIRI WFSS. We specifically target fields in COSMOS that have archival NIRISS and NIRCам grism data as well as a treasure trove of optical-to-MIR imaging and spectroscopic data providing redshifts and galaxy physical properties. We will perform an unbiased search for PAH-3.3 micron emission for >100 galaxies over $0.5 < z < 2.3$ and calibrate the PAH-3.3 emission as a star-formation indicator directly at cosmic noon. We will use diagnostic nebular emission lines from archival grism data to control for dependencies on dust, metallicity, and AGN contamination. We will also quantify the correlation between PAH emission and the 2175Å UV-bump as well as burstiness in star-formation.

Lastly, these data will also enable the first blind search for H α emitters at $z \sim 7$.

OBSERVING DESCRIPTION

We will use MIRI/WFSS to obtain slitless spectroscopy observations over 2 NIRISS fields in COSMOS to measure the PAH-3.3 micron emission in >100 galaxies at cosmic noon and calibrate it as a SF indicator. We will use the redshifts measured from the optical lines in the NIRISS grism data to identify the PAH-3.3 emission. We will also use the full suite of multi-wavelength photometry and spectroscopy for our analysis.

Given the NIRISS field-of-view, we will tile over a 1x2 mosaic with 15% overlap to match the on-sky area coverage at best as possible. We use the NIRISS images to fine tune the exact pointing. We will use the P750L disperser for our grism images. The corresponding direct images will be taken with the F560W filter. We require ~ 8.7 hrs to reach the required SNR for our science objectives. We adhere to the recommended limit of 100 groups for MIRI. The total exposure time is split over 4 dithered pointings with the cycling dither pattern and a medium pattern size. Given 2 fields (with 2

MIRI pointings each) and accounting for overheads in the APT, we estimate a total charged time of 43.1 hrs.

Proposal 11605 - Targets - Calibrating 3.3 micron PAH feature as a star-formation indicator at cosmic noon

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Par028	RA: 10 00 21.0100 (150.0875417d) Dec: +02 25 6.40 (2.41844d) Equinox: J2000		
	Comments: Category=Galaxy Description=[High-redshift galaxies]				
Fixed Targets	(2)	Par052	RA: 10 00 37.1700 (150.1548750d) Dec: +02 02 33.40 (2.04261d) Equinox: J2000		
	Comments: Category=Galaxy Description=[High-redshift galaxies]				

Proposal 11605 - Observation 1 - Calibrating 3.3 micron PAH feature as a star-formation indicator at cosmic noon

Observation	Proposal 11605, Observation 1: Par028											Fri Mar 13 21:08:34 GMT 2026
	Diagnostic Status: Warning											
Observing Template: MIRI Wide Field Slitless Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	Par028	RA: 10 00 21.0100 (150.0875417d) Dec: +02 25 6.40 (2.41844d) Equinox: J2000									
Comments: Category=Galaxy Description=[High-redshift galaxies]												
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	2	10.0		15.0		0.0		0.0		DEFAULT	
Dithers	#	Dither Type			Starting Point		Number of Points			Pattern Size		
	1	CYCLING			1		4			MEDIUM		
Direct Image	#	Exposure Type	Filter	Disperser	Readout Pattern	Groups/Int	Integrations/Exp	Two Extra Dithers	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	DIRECT	F560W		FASTR1	50	1	NO	1	1	138.752	
	2	DIRECT	F560W		FASTR1	50	1	NO	1	1	138.752	
Spectral Elements	#	Exposure Type	Filter	Disperser	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	WFSS		P750L	FASTR1	100	28	4	112	31380.152		

Proposal 11605 - Observation 1 - Calibrating 3.3 micron PAH feature as a star-formation indicator at cosmic noon

Special Requirements	Group Visits within 53.0 Days Aperture PA Range 9 to 29 Degrees (V3 4.16455103 to 24.16455103) Aperture PA Range 99 to 119 Degrees (V3 94.16455103 to 114.16455103) Aperture PA Range 189 to 209 Degrees (V3 184.16455103 to 204.16455103) Aperture PA Range 279 to 299 Degrees (V3 274.16455103 to 294.16455103) Visits Same PA
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Proposal 11605 - Observation 2 - Calibrating 3.3 micron PAH feature as a star-formation indicator at cosmic noon

Observation	Proposal 11605, Observation 2: Par052											Fri Mar 13 21:08:34 GMT 2026
	Diagnostic Status: Warning											
Observing Template: MIRI Wide Field Slitless Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	Par052	RA: 10 00 37.1700 (150.1548750d) Dec: +02 02 33.40 (2.04261d) Equinox: J2000									
Comments: Category=Galaxy Description=[High-redshift galaxies]												
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	2	10.0		15.0		0.0		0.0		DEFAULT	
Dithers	#	Dither Type			Starting Point		Number of Points			Pattern Size		
	1	CYCLING			1		4			MEDIUM		
Direct Image	#	Exposure Type	Filter	Disperser	Readout Pattern	Groups/Int	Integrations/Exp	Two Extra Dithers	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	DIRECT	F560W		FASTR1	50	1	NO	1	1	138.752	
	2	DIRECT	F560W		FASTR1	50	1	NO	1	1	138.752	
Spectral Elements	#	Exposure Type	Filter	Disperser	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	WFSS		P750L	FASTR1	100	28	4	112	31380.152		

Proposal 11605 - Observation 2 - Calibrating 3.3 micron PAH feature as a star-formation indicator at cosmic noon

Special Requirements	Group Visits within 53.0 Days Aperture PA Range 13 to 33 Degrees (V3 8.16455103 to 28.16455103) Aperture PA Range 103 to 123 Degrees (V3 98.16455103 to 118.16455103) Aperture PA Range 193 to 213 Degrees (V3 188.16455103 to 208.16455103) Aperture PA Range 283 to 303 Degrees (V3 278.16455103 to 298.16455103) Visits Same PA
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